Assignment 2 - Diffusion Models from Scratch
Neural Graphics (3683-8901)

Eden Avrahami (207106444), Guy Yaffe (207253980)

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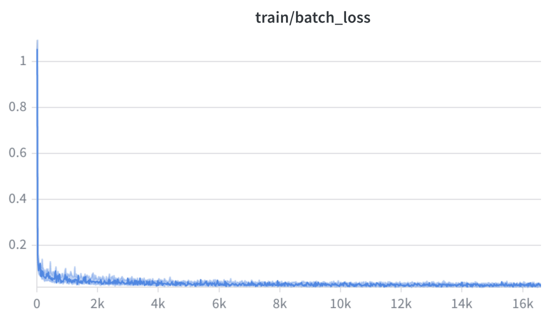
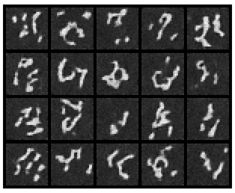
2. Unconditional Diffusion Framework



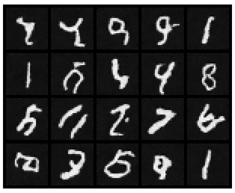
(a) Batch Loss

(b) Epoch Loss

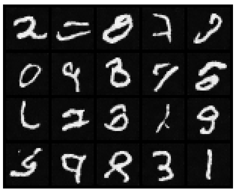
Figure 1: Training loss curves for the unconditional denoiser.



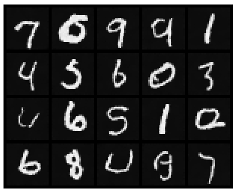
(a) Epoch 1



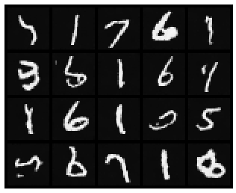
(b) Epoch 5



(c) Epoch 10



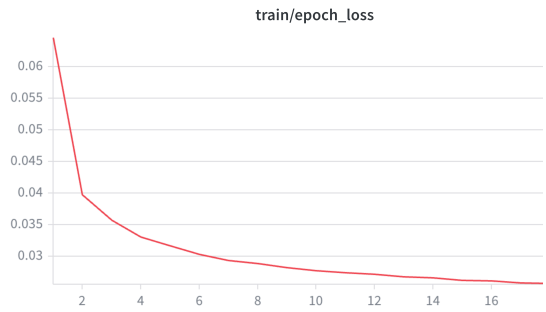
(d) Epoch 15



(e) Epoch 20

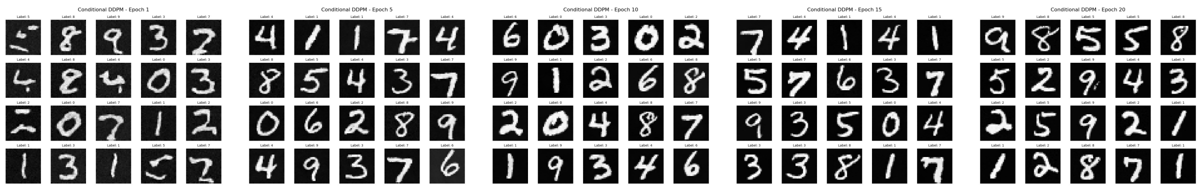
Figure 2: Evaluation samples generated by the unconditional denoiser at epochs 1, 5, 10, 15, and 20.

3. Implementing Class-Conditioned Diffusion Framework with CFG



(a) Batch Loss (b) Epoch Loss

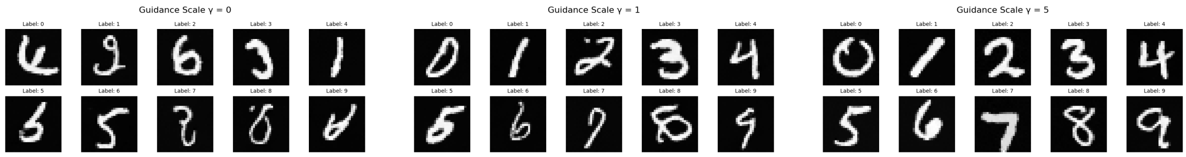
Figure 3: Training loss curves for the class-conditioned denoiser.



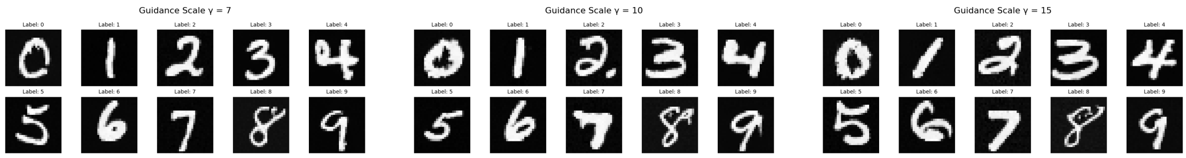
(a) Epoch 1 (b) Epoch 5 (c) Epoch 10 (d) Epoch 15 (e) Epoch 20

Figure 4: Evaluation samples generated by the class-conditioned denoiser at epochs 1, 5, 10, 15, and 20, using a guidance scale of $\gamma = 5.0$.

3.4 Experimenting with Different Guidance Scales



(a) $\gamma = 0$ (b) $\gamma = 1$ (c) $\gamma = 5$



(d) $\gamma = 7$ (e) $\gamma = 10$ (f) $\gamma = 15$

Figure 5: Generated samples for digits 0-9 across varying CFG scales ($\gamma \in \{0, 1, 5, 7, 10, 15\}$).

Analysis of Visual Effects

The results demonstrate a direct trade-off between sample realism and prompt compliance: lower guidance scales produce images more native to the dataset, while higher scales force strict adherence to the condition.

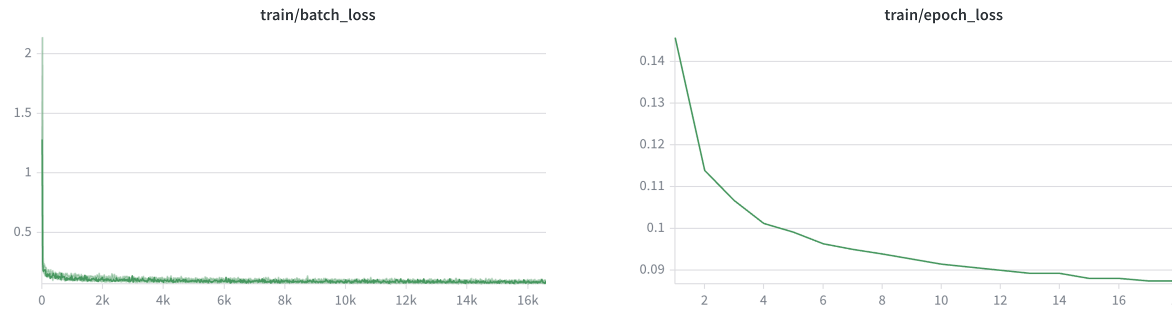
Low Guidance ($\gamma \in \{0, 1\}$): At $\gamma = 0$, the model completely ignores the conditioning label. Digits fit the dataset visually, however, are completely random (e.g., generating a '9' when asked for a '1'). At

$\gamma = 1$, the model attempts to incorporate the condition, but the adherence is weak, resulting in ambiguous or malformed shapes (e.g., the '8' at $\gamma = 1$). Mathematically, at these low scales, the generation process relies entirely on or heavily favors the unconditional prior (ϵ_{uncond}). The model prioritizes generating a sample that fits within the distribution of the training data over obeying the prompt.

High Guidance ($\gamma \in \{10, 15\}$): In contrast, at high scales like $\gamma = 10$ and particularly $\gamma = 15$, the model complies to the requested class label, but the structure of the images differs than the original dataset. For example, strokes become overly thick and the digit takes up more space in the image. By heavily amplifying the direction vector ($\epsilon_{cond} - \epsilon_{uncond}$), we force the sampling process outside the learned manifold of the dataset images, and focus on enhancing features of the conditioned digit.

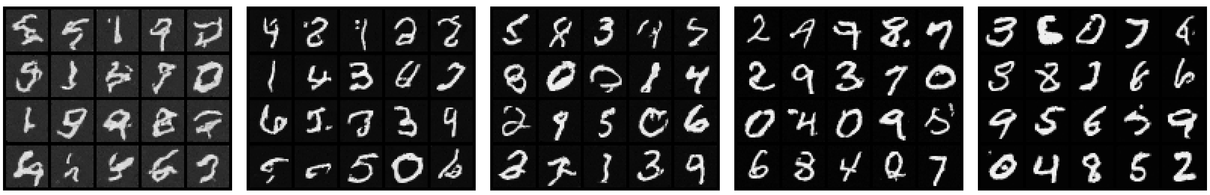
The Middle Ground ($\gamma \in \{5, 7\}$): Here we see a nice balance of clean images that fit the dataset, where each digit can be easily identified as its conditioned prompt. This suggests that these values provide a good balance of the tradeoff between ϵ_{uncond} and $\epsilon_{cond} - \epsilon_{uncond}$.

4. Flow Matching (Bonus)



(a) Batch Loss (b) Epoch Loss

Figure 6: Training loss curves for the flow-matching model.



(a) Epoch 1 (b) Epoch 5 (c) Epoch 10 (d) Epoch 15 (e) Epoch 20

Figure 7: Evaluation samples generated by the flow-matching model at epochs 1, 5, 10, 15, and 20.